

AMRITAA SHANKAR

AI & Embedded Systems Engineer

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PROFILE

AI and software engineering undergraduate (3.7/4.0) with hands-on experience building anomaly detection models on real industrial sensor data, IoT systems with embedded hardware (ESP32), and full-stack applications that turn raw signals into decisions people can act on. Comfortable across the stack from data engineering and ML through to the dashboards and interfaces built on top. Interested in how AI and applied engineering intersect with security: detecting abnormal patterns in systems, securing connected devices, and building tools people can trust.

SKILLS

AI / ML & Anomaly Detection TensorFlow, Scikit-learn, XGBoost, CNNs, LSTMs, time-series anomaly detection, LLM integration (OpenAI API)

IoT & Embedded Systems ESP32, sensor integration, hardware-software system design

Languages & Frameworks Python, C, C++, JavaScript, React, Node.js, SQL/PostgreSQL, MATLAB

Tools & Platforms Git/GitHub, Power BI, Power Automate & Power Apps, SAP Cloud ERP

WORK EXPERIENCE

Digital & AI Intern, Shared Services — ADNOC Group / Al Dhafra Petroleum, UAE. Dec 2025 – Feb 2026

- Built ML-based anomaly detection models for predictive maintenance on high-frequency industrial time-series sensor data — identifying abnormal operating patterns before failure.
- Engineered lag/temporal features and trained models with Scikit-learn and XGBoost to flag anomalous behavior across pump and reservoir systems.
- Built Power BI dashboards visualizing anomaly scores and operational risk indicators for engineers without a data science background.

AI & Web Development Intern — ASPIR, USA Jul 2025 – Sep 2025

Technical Lead, Frontend Team

- Led development of a full-featured React.js web platform with a responsive, component-based UI for internal operations and user access.
- Engineered an OpenAI LLM-powered chatbot, integrating backend APIs and modular UI components for scalability and real-time performance.

PROJECTS

Space Debris Detection & Trajectory Prediction AI. 2025

Samsung Innovation Campus

- Trained CNNs for debris identification and LSTMs for orbital trajectory forecasting — detecting and classifying anomalous objects to support satellite safety.

PowerSprout — IoT-Enabled Microbial Fuel Cell. 2025

University of Wollongong in Dubai · 3rd Place, UOWD Innovation Fair 2025

- Designed and built an ESP32-based connected hardware system with real-time sensors, a Flask ML model, and a full-stack monitoring dashboard.

SilahGrid — AI-Powered Smart Edge Cooling Network. 2025

Zayed University Innovation Hackathon 2025

- Built a distributed edge-AI system with ML-based task routing across multiple connected edge nodes, with a focus on stable, high-speed operation.

EDUCATION

Bachelor of Engineering (Honors), Computer and Autonomous Systems Engineering. 2023 – 2027

University of Wollongong in Dubai · GPA 3.7/4.0

PUBLICATIONS & HONORS

- Best Paper Award, ICEEIE 2025 — "Comparative Study of RF Shielding Using SAR Simulations Highlighting the Effectiveness of Nanocomposite Materials" — applied research on electromagnetic protection.
- Certifications: Full Stack Web Dev (Udemy) · AI Course (Samsung Innovation Campus) · Power BI (Udemy) · SAP Cloud ERP.